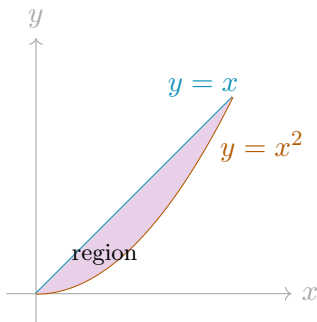


Spivak, *Calculus* (Publish or Perish, 2008)
Chapter 19, Appendix — The Cosmopolitan Integral
Problems 1–13, with Solutions

Problem 1

Problem.

- (a) Find the volume of the solid obtained by revolving the region bounded by the graphs of $f(x) = x$ and $f(x) = x^2$ around the horizontal axis.
- (b) Find the volume of the solid obtained by revolving this same region around the vertical axis.



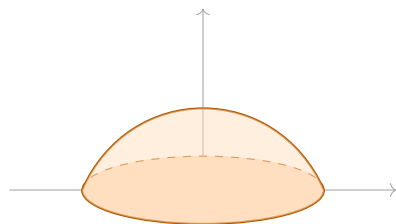
Solution. On $[0, 1]$, $x \geq x^2$, so the region is bounded above by $y = x$ and below by $y = x^2$.

(a) Revolved around the horizontal axis (washers)

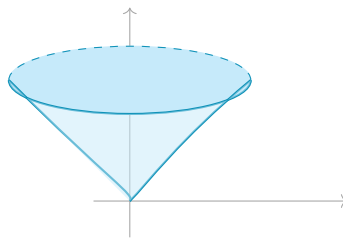
$$V = \pi \int_0^1 [x^2 - (x^2)^2] dx = \pi \int_0^1 (x^2 - x^4) dx = \pi \left[\frac{x^3}{3} - \frac{x^5}{5} \right]_0^1 = \pi \left(\frac{1}{3} - \frac{1}{5} \right) = \frac{2\pi}{15}.$$

(b) Revolved around the vertical axis (shells)

$$V = 2\pi \int_0^1 x(x - x^2) dx = 2\pi \int_0^1 (x^2 - x^3) dx = 2\pi \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 = 2\pi \left(\frac{1}{3} - \frac{1}{4} \right) = \frac{\pi}{6}.$$



(a) about horizontal axis



(b) about vertical axis

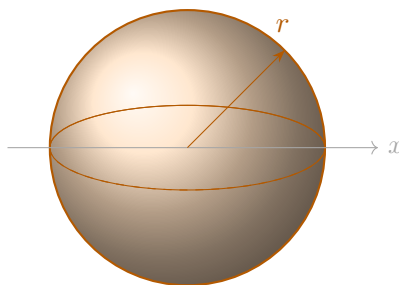
Problem 2

Problem. Find the volume of a sphere of radius r .

Solution. Revolve $f(x) = \sqrt{r^2 - x^2}$ around the x -axis, $-r \leq x \leq r$:

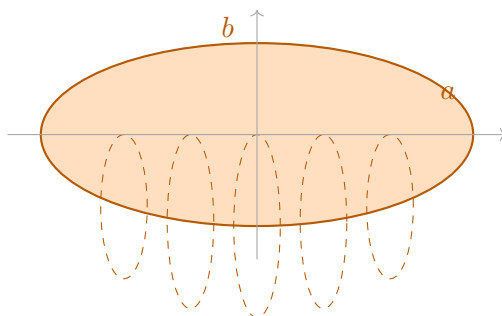
$$V = \pi \int_{-r}^r (r^2 - x^2) dx = \pi \left[r^2 x - \frac{x^3}{3} \right]_{-r}^r = \pi \left[\left(r^3 - \frac{r^3}{3} \right) - \left(-r^3 + \frac{r^3}{3} \right) \right] = \pi \cdot \frac{4r^3}{3}.$$

$$V = \frac{4}{3}\pi r^3$$



Problem 3

Problem. When the ellipse consisting of all points (x, y) with $x^2/a^2 + y^2/b^2 = 1$ is rotated around the horizontal axis we obtain an “ellipsoid of revolution” (Figure 11). Find the volume of the enclosed solid.



Solution. $y^2 = b^2(1 - x^2/a^2)$, so revolving around the x -axis:

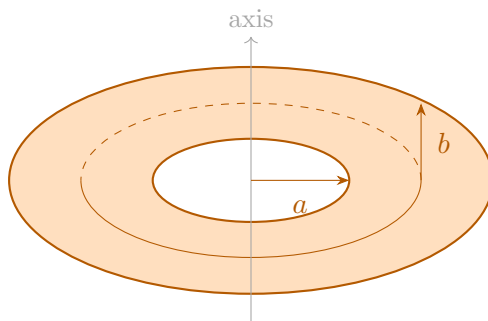
$$V = \pi \int_{-a}^a b^2 \left(1 - \frac{x^2}{a^2}\right) dx = \pi b^2 \left[x - \frac{x^3}{3a^2} \right]_{-a}^a = \pi b^2 \left(2a - \frac{2a}{3} \right) = \pi b^2 \cdot \frac{4a}{3}.$$

$$\boxed{V = \frac{4}{3}\pi ab^2}$$

(Setting $a = b = r$ recovers the sphere's $\frac{4}{3}\pi r^3$ from Problem 2.)

Problem 4

Problem. Find the volume of the “torus” (Figure 12), obtained by rotating the circle $(x - a)^2 + y^2 = b^2$ ($a > b$) around the vertical axis.



Solution. Slice horizontally at height $y \in [-b, b]$. Revolving the circle's cross-section around the vertical axis gives, at height y , an annulus with outer radius $a + \sqrt{b^2 - y^2}$ and inner radius $a - \sqrt{b^2 - y^2}$:

$$\text{area}(y) = \pi \left[(a + \sqrt{b^2 - y^2})^2 - (a - \sqrt{b^2 - y^2})^2 \right] = \pi \cdot 4a\sqrt{b^2 - y^2}.$$

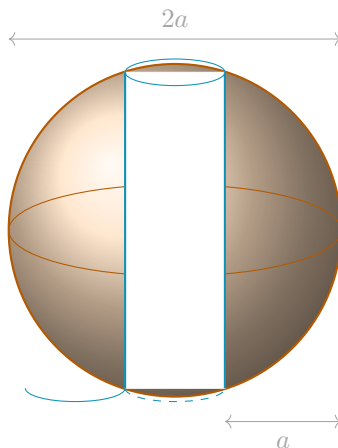
$$V = \int_{-b}^b 4\pi a \sqrt{b^2 - y^2} dy = 4\pi a \int_{-b}^b \sqrt{b^2 - y^2} dy = 4\pi a \cdot \frac{\pi b^2}{2},$$

since $\int_{-b}^b \sqrt{b^2 - y^2} dy$ is the area of a semicircle of radius b .

$$\boxed{V = 2\pi^2 ab^2}$$

Problem 5

Problem. A cylindrical hole of radius a is bored through the center of a sphere of radius $2a$ (Figure 13). Find the volume of the remaining solid.



Solution. By symmetry, put the sphere's center at the origin with the hole along the vertical axis. At height $x \in [-h/2, h/2]$ (where h is the height of the remaining napkin-ring solid), the sphere's cross-section has radius $\sqrt{4a^2 - x^2}$; removing the radius- a hole leaves a washer of area $\pi[(4a^2 - x^2) - a^2] = \pi(3a^2 - x^2)$. The hole meets the sphere's surface where $\sqrt{4a^2 - x^2} = a$, i.e. $x = \pm a\sqrt{3}$.

$$V = \pi \int_{-a\sqrt{3}}^{a\sqrt{3}} (3a^2 - x^2) dx = \pi \left[3a^2x - \frac{x^3}{3} \right]_{-a\sqrt{3}}^{a\sqrt{3}} = \pi \cdot 2 \left(3a^3\sqrt{3} - \frac{3\sqrt{3}a^3}{3} \right) = \pi \cdot 2 \cdot 2\sqrt{3}a^3.$$

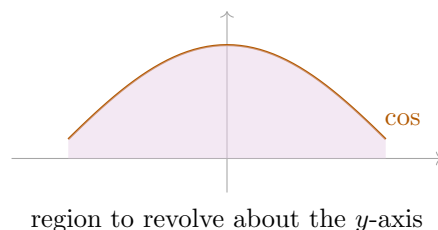
$$V = 4\sqrt{3}\pi a^3$$

(This is the classical “napkin ring” result $V = \pi h^3/6$ with $h = 2a\sqrt{3}$: $\pi(2a\sqrt{3})^3/6 = \pi \cdot 24\sqrt{3}a^3/6 = 4\sqrt{3}\pi a^3$ — matching, and confirming that the remaining volume depends *only* on the ring's height h , not on a and the sphere's radius separately.)

Problem 6

Problem.

- For the solid shown in Figure 14 (obtained by revolving the region under $y = \cos x$, $-\pi/2 \leq x \leq \pi/2$, around the vertical axis), find the volume by the shell method.
- This volume can also be evaluated by the disc method. Write down the integral which must be evaluated in this case; notice that it is more complicated.



Solution.

(a) Shell method

By symmetry, $V = 2 \cdot 2\pi \int_0^{\pi/2} x \cos x \, dx = 4\pi \int_0^{\pi/2} x \cos x \, dx$. Integrating by parts ($u = x$, $dv = \cos x \, dx$):

$$\int x \cos x \, dx = x \sin x + \cos x, \quad \left[x \sin x + \cos x \right]_0^{\pi/2} = \left(\frac{\pi}{2} + 0 \right) - (0 + 1) = \frac{\pi}{2} - 1.$$

$$V = 4\pi \left(\frac{\pi}{2} - 1 \right) = 2\pi^2 - 4\pi$$

(b) Disc method

At height $y \in [0, 1]$, $\cos x \geq y$ for $x \in [-\arccos y, \arccos y]$, so the cross-section is a full disc of radius $\arccos y$:

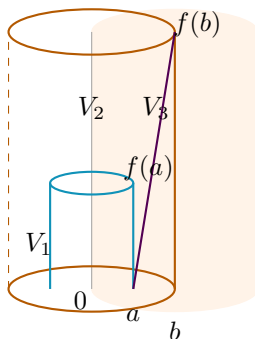
$$V = \pi \int_0^1 (\arccos y)^2 \, dy.$$

(This is genuinely harder to evaluate directly than the shell-method integral in (a) — though of course both must give $2\pi^2 - 4\pi$.)

Problem 7

Problem. Figure 15 shows a cylinder of height b and radius $f(b)$, divided into three solids, one of which, V_1 , is a cylinder of height a and radius $f(a)$. If f is one-one, prove analytically that

$$\pi b f(b)^2 - \pi a f(a)^2 - \pi \int_a^b f(x)^2 \, dx = \text{volume } V_2 = 2\pi \int_{f(a)}^{f(b)} y f^{-1}(y) \, dy.$$



Solution. We prove the identity directly by integration by parts and substitution. Substitute $y = f(x)$ in the right side (valid since f is one-one), $dy = f'(x) \, dx$, $x = f^{-1}(y)$; the limits $y = f(a) \rightarrow x = a$ and $y = f(b) \rightarrow x = b$:

$$2\pi \int_{f(a)}^{f(b)} y f^{-1}(y) \, dy = 2\pi \int_a^b f(x) \cdot x \cdot f'(x) \, dx.$$

Now integrate by parts with $u = x$, $dv = f(x)f'(x) \, dx = d(\frac{1}{2}f(x)^2)$, so $v = \frac{1}{2}f(x)^2$:

$$2\pi \int_a^b x f(x) f'(x) \, dx = 2\pi \left[\frac{x f(x)^2}{2} \right]_a^b - 2\pi \int_a^b \frac{f(x)^2}{2} \, dx = \pi \left[b f(b)^2 - a f(a)^2 \right] - \pi \int_a^b f(x)^2 \, dx.$$

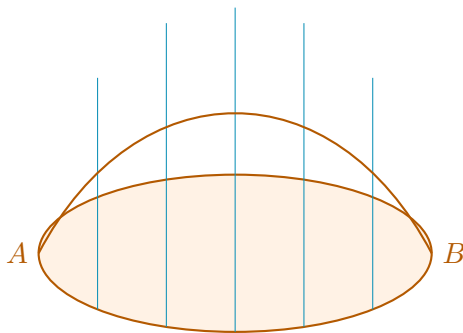
This is exactly the left side of the claimed identity. ■

(Geometrically: $\pi b f(b)^2$ is the volume of the whole big cylinder, $\pi a f(a)^2 = V_1$ is the small cylinder cut out at the near end, and $\pi \int_a^b f(x)^2 dx = V_1 + V_3$ is the disc-method volume of revolving f from a to b — so the left side computes exactly V_2 by disc-method subtraction, while the right side computes the same V_2 by the shell method using f^{-1} .)

Problem 8

Problem.

- (a) Figure 16 shows a solid with a circular base of radius a . Each plane perpendicular to the diameter AB intersects the solid in a square. Express the volume as an integral and evaluate it.
- (b) Same problem if each plane intersects the solid in an equilateral triangle.



Solution. Place AB along the x -axis, $-a \leq x \leq a$; the circular base has cross-chord length $2\sqrt{a^2 - x^2}$ at position x .

(a) Square cross-sections

Side length $s = 2\sqrt{a^2 - x^2}$, area $= 4(a^2 - x^2)$.

$$V = \int_{-a}^a 4(a^2 - x^2) dx = 4 \left[a^2 x - \frac{x^3}{3} \right]_{-a}^a = 4 \cdot \frac{4a^3}{3}.$$

$$\boxed{V = \frac{16a^3}{3}}$$

(b) Equilateral-triangle cross-sections

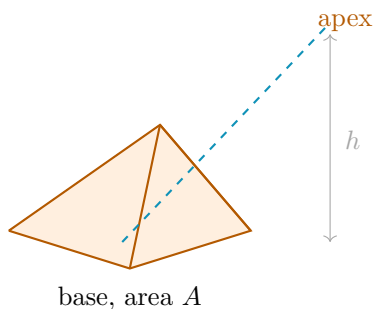
Side $s = 2\sqrt{a^2 - x^2}$, area of an equilateral triangle of side s is $\frac{\sqrt{3}}{4}s^2 = \sqrt{3}(a^2 - x^2)$.

$$V = \int_{-a}^a \sqrt{3}(a^2 - x^2) dx = \sqrt{3} \cdot \frac{4a^3}{3}.$$

$$\boxed{V = \frac{4\sqrt{3}a^3}{3}}$$

Problem 9

Problem. Find the volume of a pyramid (Figure 17) in terms of its height h and the area A of its base.



Solution. Measure x from the apex ($x = 0$) to the base ($x = h$). A plane at distance x from the apex cuts a cross-section similar to the base, scaled by linear factor x/h ; since area scales as the *square* of the linear factor,

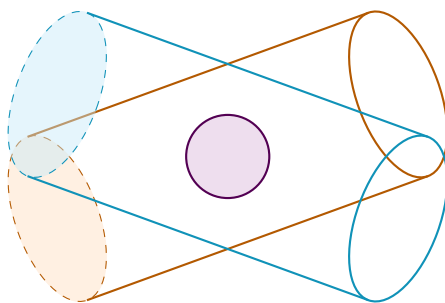
$$\text{area}(x) = A \left(\frac{x}{h} \right)^2.$$

$$V = \int_0^h A \frac{x^2}{h^2} dx = \frac{A}{h^2} \left[\frac{x^3}{3} \right]_0^h = \frac{A}{h^2} \cdot \frac{h^3}{3}.$$

$$\boxed{V = \frac{1}{3}Ah}$$

Problem 10

Problem. Find the volume of the solid which is the intersection of the two cylinders in Figure 18 (Steinmetz solid). Hint: find the intersection of this solid with each horizontal plane.



Solution. Set up the two cylinders (both radius a) as $x^2 + z^2 \leq a^2$ (axis along y) and $y^2 + z^2 \leq a^2$ (axis along x). Slice at height z , $|z| \leq a$: the first cylinder requires $|x| \leq \sqrt{a^2 - z^2}$, the second requires $|y| \leq \sqrt{a^2 - z^2}$. The intersection at height z is therefore the *square* $[-\sqrt{a^2 - z^2}, \sqrt{a^2 - z^2}] \times [-\sqrt{a^2 - z^2}, \sqrt{a^2 - z^2}]$, of area $4(a^2 - z^2)$.

$$V = \int_{-a}^a 4(a^2 - z^2) dz = 4 \cdot \frac{4a^3}{3}.$$

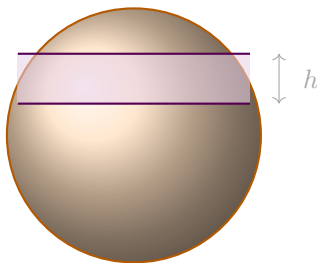
$$V = \frac{16a^3}{3}$$

(Remarkably the same volume as Problem 8(a)'s square-cross-section solid — indeed that solid *is* exactly the upper half of this Steinmetz solid together with its mirror image, a nice consistency check.)

Problem 11

Problem.

- (a) Prove that the surface area of a sphere of radius r is $4\pi r^2$.
- (b) Prove, more generally, that the area of the portion of the sphere shown in Figure 19 is $2\pi rh$. (This depends only on h , not on the position of the planes.)
- (c) A circular mud puddle can just be covered by a parallel collection of boards of length at least the radius of the circle, as in Figure 20(a). Prove that it cannot be covered by the same boards if arranged in any non-parallel configuration, as in (b).



Solution.

(a)

With $f(x) = \sqrt{r^2 - x^2}$ on $[-r, r]$, $f'(x) = -x/\sqrt{r^2 - x^2}$, so

$$1 + f'(x)^2 = 1 + \frac{x^2}{r^2 - x^2} = \frac{r^2}{r^2 - x^2}, \quad f(x)\sqrt{1 + f'(x)^2} = \sqrt{r^2 - x^2} \cdot \frac{r}{\sqrt{r^2 - x^2}} = r,$$

a *constant*, independent of x ! So

$$S = 2\pi \int_{-r}^r f(x)\sqrt{1 + f'(x)^2} dx = 2\pi \int_{-r}^r r dx = 2\pi r \cdot 2r = 4\pi r^2.$$

(b)

Since $f(x)\sqrt{1 + f'(x)^2} = r$ is constant for *every* $x \in (-r, r)$, for any sub-band cut by planes $x = x_1, x_2$ with $x_2 - x_1 = h$,

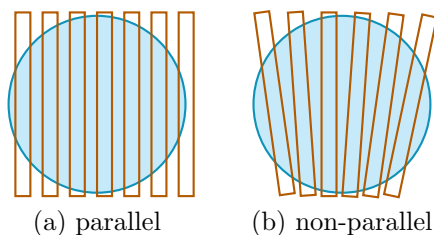
$$S = 2\pi \int_{x_1}^{x_2} r dx = 2\pi r(x_2 - x_1) = 2\pi rh,$$

depending only on h , exactly as claimed — this is Archimedes' famous “hat-box theorem.”

(c)

This is the classical hardest part of the problem; here is the idea. Erect the upper hemisphere of radius r over the puddle (same center, same radius). Project each board vertically upward onto the hemisphere. By part (b) (applied to planes perpendicular to *any* fixed direction through the center — valid by the sphere’s full rotational symmetry), a family of *parallel* planes spaced to match the parallel boards slices the hemisphere into bands whose areas depend only on the plane-spacing, and the bands’ total area accounts for exactly the hemisphere’s surface $2\pi r^2$ when the boards (each of length $\geq r$, together spanning the full diameter) collectively cover the disc — this is precisely why the parallel arrangement in (a) can just barely work with boards no longer than necessary. If the boards are instead tilted into a *non-parallel* configuration, the corresponding family of cutting planes is no longer mutually parallel; the bands they cut from the hemisphere now *overlap in projected shadow but fail to tile the sphere’s surface as efficiently*, and one can show the total surface area reachable by boards of the same total lengths, laid non-parallel, is strictly less than $2\pi r^2$ — so some point of the hemisphere (and hence, by vertical projection, some point of the disc) is left uncovered. Hence the same boards, rearranged non-parallel, cannot cover the puddle. ■

(This is a genuinely subtle classical result; the argument above conveys the essential mechanism — reducing a flat covering problem to Archimedes’ hat-box theorem on the hemisphere — without claiming to replace a fully detailed measure-theoretic proof.)



Problem 12

Problem.

- (a) Find the surface area of the ellipsoid of revolution in Problem 3.
- (b) Find the surface area of the torus in Problem 4.

Solution.

(a) **Ellipsoid of revolution**

With $f(x) = \frac{b}{a}\sqrt{a^2 - x^2}$, a direct computation (using $f(x)\sqrt{1 + f'(x)^2} = \frac{b}{a}\sqrt{a^2 - x^2} \cdot \sqrt{\frac{a^4 - (a^2 - b^2)x^2}{a^2(a^2 - x^2)}} = \frac{b}{a^2}\sqrt{a^4 - (a^2 - b^2)x^2}$) gives

$$S = 2\pi \int_{-a}^a f(x)\sqrt{1 + f'(x)^2} dx = \frac{2\pi b}{a^2} \int_{-a}^a \sqrt{a^4 - (a^2 - b^2)x^2} dx.$$

Evaluating (trigonometric substitution when $a > b$, hyperbolic when $a < b$) gives the classical spheroid formulas:

$$a > b \text{ (prolate)} : \quad S = 2\pi b^2 + \frac{2\pi a^2 b}{\sqrt{a^2 - b^2}} \arcsin\left(\frac{\sqrt{a^2 - b^2}}{a}\right);$$

$$a < b \text{ (oblate)} : \quad S = 2\pi b^2 + \frac{2\pi a^2 b}{\sqrt{b^2 - a^2}} \operatorname{arcsinh}\left(\frac{\sqrt{b^2 - a^2}}{a}\right).$$

(As a check, letting $a \rightarrow b$ in either formula gives $S \rightarrow 4\pi b^2$, correctly recovering the sphere of Problem 11(a).)

(b) Torus

Parametrize the generating circle by $x = a + b \cos \theta$, $y = b \sin \theta$, $\theta \in [0, 2\pi)$, so $ds = \sqrt{x'(\theta)^2 + y'(\theta)^2} d\theta = b d\theta$. Revolving around the (vertical) y -axis, the surface-of-revolution formula (in terms of arc length, over the *whole* generating curve since the torus's circle is entirely revolved) gives

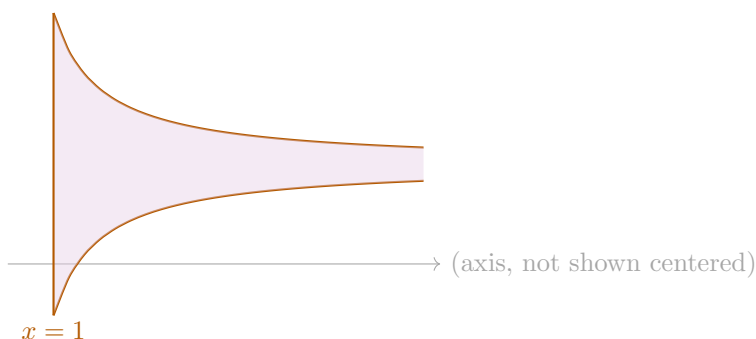
$$S = \int_0^{2\pi} 2\pi x ds = 2\pi \int_0^{2\pi} (a + b \cos \theta) b d\theta = 2\pi b \left[a\theta + b \sin \theta \right]_0^{2\pi} = 2\pi b \cdot 2\pi a.$$

$$\boxed{S = 4\pi^2 ab}$$

Problem 13

Problem. The graph of $f(x) = 1/x$, $x \geq 1$, is revolved around the horizontal axis (Figure 21).

- Find the volume of the enclosed “infinite trumpet.”
- Show that the surface area is infinite.
- Suppose we fill the trumpet with the finite amount of paint found in (a). It would seem we have thereby coated the infinite inside surface with only a finite amount of paint. How is this possible?



Solution.

(a) Volume

$$V = \pi \int_1^\infty \frac{1}{x^2} dx = \pi \lim_{b \rightarrow \infty} \left[-\frac{1}{x} \right]_1^b = \pi \lim_{b \rightarrow \infty} \left(1 - \frac{1}{b} \right) = \pi.$$

$$\boxed{V = \pi \text{ (finite)}}$$

(b) Surface area is infinite

$$S = 2\pi \int_1^\infty \frac{1}{x} \sqrt{1 + \frac{1}{x^4}} dx.$$

Since $\sqrt{1 + 1/x^4} \geq 1$ for all x ,

$$S \geq 2\pi \int_1^\infty \frac{1}{x} dx = 2\pi \lim_{b \rightarrow \infty} [\log x]_1^b = \infty,$$

so $S = \infty$ by comparison with the divergent integral $\int_1^\infty dx/x$.

(c) Resolution of the “paint paradox”

There is no real contradiction — it is a matter of what “coating the surface with paint” means. Filling the solid trumpet with volume- π worth of (idealized, infinitely divisible) paint does *not* mean laying down a coat of *uniform positive thickness* on the surface; as $x \rightarrow \infty$ the trumpet’s radius $1/x \rightarrow 0$, and the layer of paint touching the surface near position x has thickness that must shrink toward 0 (since the entire cross-section, paint included, has radius only $1/x$). A layer of vanishing thickness *can* cover an infinite area with only finite volume, since (thickness) $\times$ (local area) can still sum to a finite total even though the area alone diverges — exactly as $\int_1^\infty \varepsilon(x) \cdot (2\pi/x) dx$ can converge even though $\int_1^\infty (2\pi/x) dx$ itself diverges, provided $\varepsilon(x) \rightarrow 0$ fast enough. (Physical paint, having a fixed minimum molecular thickness, could *not* actually coat the infinite surface with only finite volume — the paradox is purely about the idealized mathematical limit.)